

3.8 For water (H_2O), locate each of the following states on sketches of the T - v , p - v , and phase diagrams.

(a) $T = 300^\circ\text{F}$, $p = 20 \text{ lbf/in.}^2$

(b) $T = 300^\circ\text{F}$, $p = 90 \text{ lbf/in.}^2$

(c) $T = 300^\circ\text{F}$, $v = 5 \text{ ft}^3/\text{lb}$.

KNOWN: H_2O at specified states.

FIND: Locate each specified state on sketches of the T - v , p - v , and phase diagrams.

SCHEMATIC AND GIVEN DATA:

<u>State (a)</u>	<u>State (b)</u>	<u>State (c)</u>
$T = 300^\circ\text{F}$	$T = 300^\circ\text{F}$	$T = 300^\circ\text{F}$
$p = 20 \text{ lbf/in.}^2$	$p = 90 \text{ lbf/in.}^2$	$v = 5 \text{ ft}^3/\text{lb}$

ENGINEERING MODEL:

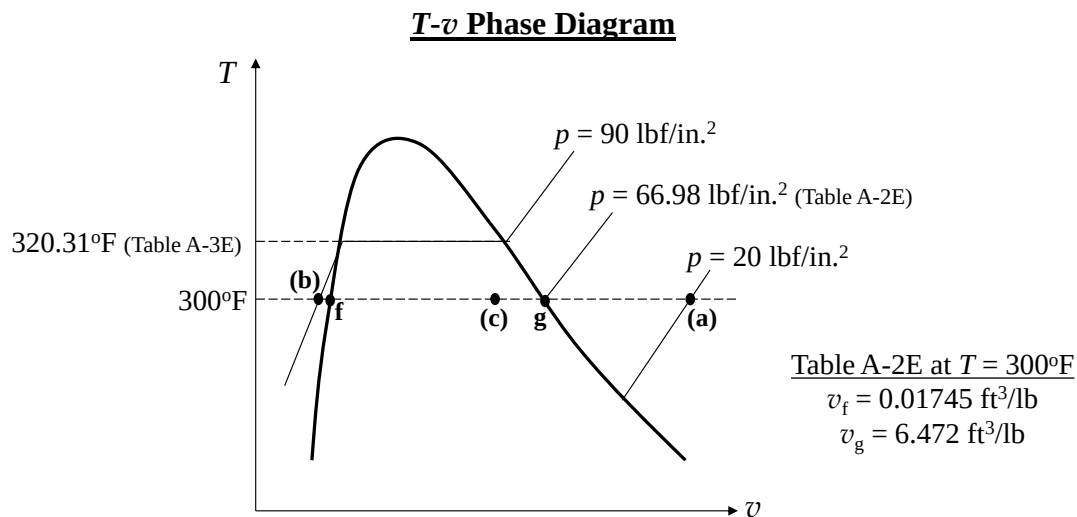
1. States (a), (b), and (c) are equilibrium states.

ANALYSIS:

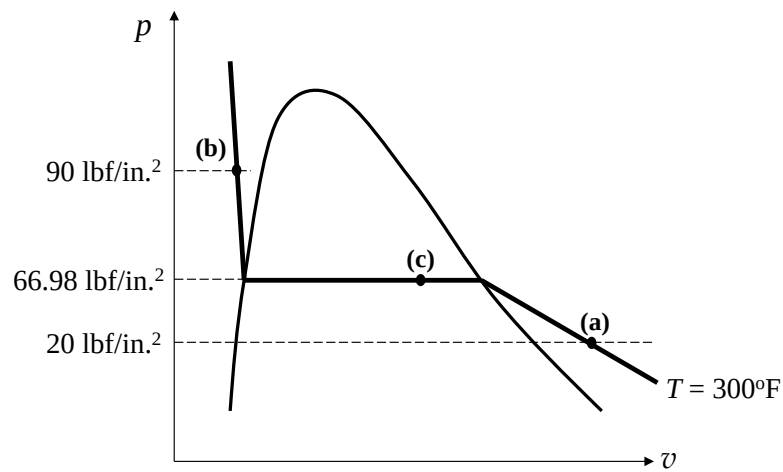
State (a): Since $p < p_{\text{sat}}$ (66.98 lbf/in.^2 from Table A-2E at $T = 300^\circ\text{F}$), state (a) is superheated vapor.

State (b): Since $p > p_{\text{sat}}$ (66.98 lbf/in.^2 from Table A-2E at $T = 300^\circ\text{F}$), state (b) is compressed liquid.

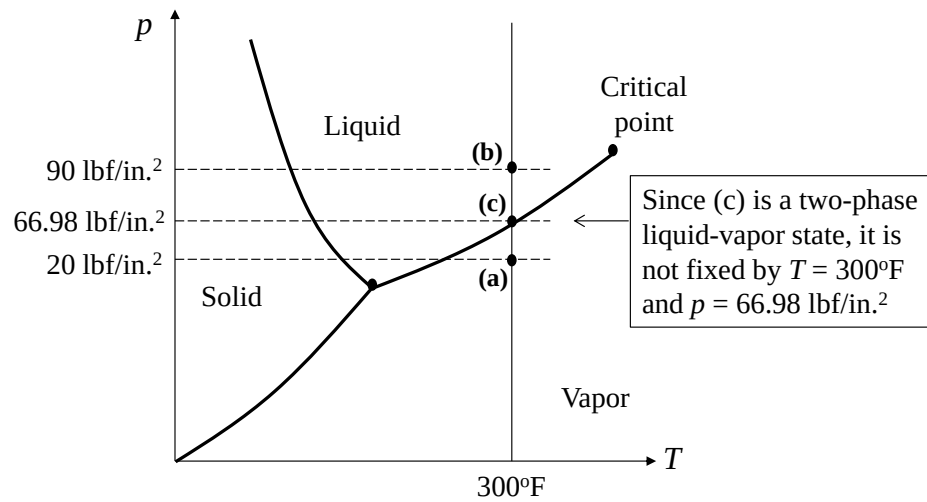
State (c): From Table A-2E at $T = 300^\circ\text{F}$, $v_f = 0.01745 \text{ ft}^3/\text{lb}$ and $v_g = 6.472 \text{ ft}^3/\text{lb}$. Since $v_f < v < v_g$, state (c) is a two-phase liquid-vapor mixture.



p - v Phase Diagram

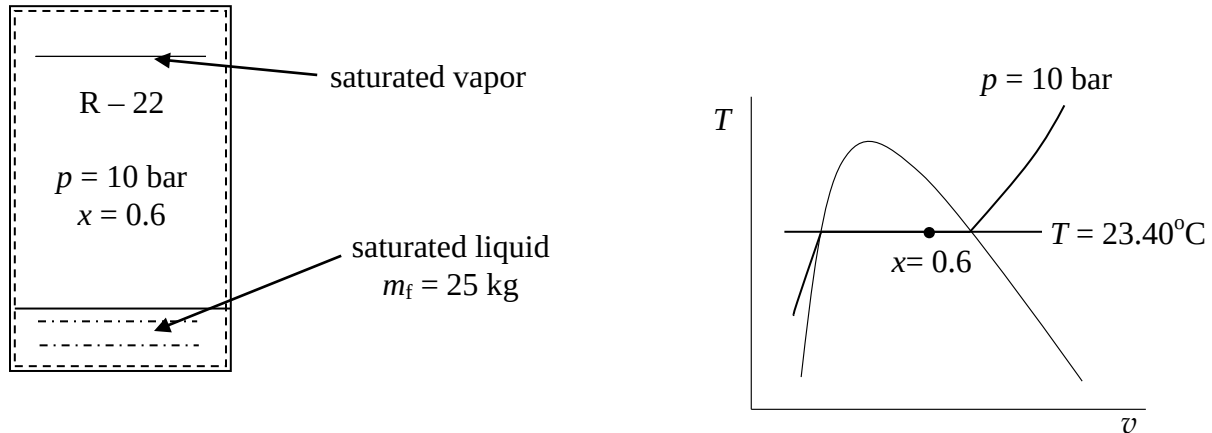


Phase Diagram



PROBLEM 3.12

A tank contains a two-phase liquid-vapor mixture of Refrigerant 22 at 10 bar. The mass of saturated liquid in the tank is 25 kg and the quality is 60%. Determine the volume of the tank, in m^3 , and the fraction of the total volume occupied by saturated vapor.



First, determine the specific volume using Eq. 3.2 and data from Table A-8 at 10 bar.

$$v_x = v_x + x(v_g - v_f) = 0.8352 \times 10^{-3} + (0.6)(0.0236 - 0.8352 \times 10^{-3}) = 0.01449 \text{ m}^3/\text{kg}$$

The total mass is determined from the mass of saturated liquid and the definition of quality, as follows.

$$m = m_f / (1 - x) = (25 \text{ kg}) / (1 - 0.6) = 62.5 \text{ kg}$$

Now, the volume is

$$V = v_x m = (0.01449 \text{ m}^3/\text{kg})(62.5 \text{ kg}) = 0.9056 \text{ m}^3$$

The volume occupied by saturated vapor is

$$V_g = v_g(m - m_f) = (0.0236 \text{ m}^3/\text{kg})(62.5 - 25)\text{kg} = 0.885 \text{ m}^3$$

$$\text{fraction occupied by vapor} = (0.885)/(0.9056) = 0.977 \text{ (97.7\%)}$$

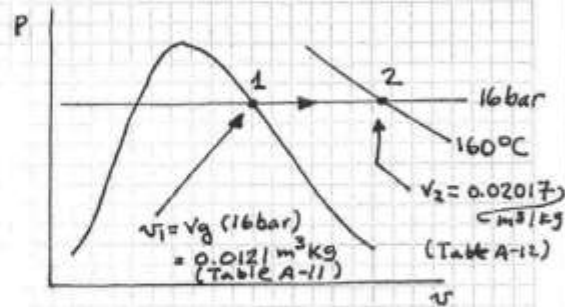
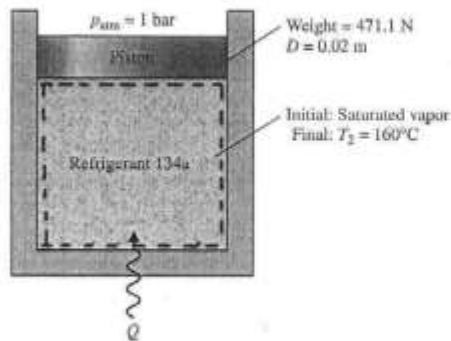
Note: Even though the vapor is only 60% of the mixture by mass, it occupies nearly the entire volume because the specific volume of saturated liquid is much smaller than the specific volume of saturated vapor at this pressure.

PROBLEM 3.25

KNOWN: Refrigerant 134a contained in a piston-cylinder assembly is slowly heated. Data is provided at the initial and final states.

FIND: Determine the work, in kJ/kg.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The Refrigerant 134a is the closed system.
2. During the process the refrigerant is slowly heated and the piston moves smoothly in the cylinder.
3. Volume change is the only work mode.

ANALYSIS

With assumptions 2 and 3 the work can be evaluated from Eq. 2.17, expressed as $W/m = \int_1^2 p dv$. To apply this requires that states 1 and 2 be fixed and pressure-volume relation be determined.

Using a force balance for the piston, we conclude the pressure of the refrigerant remains constant during the process. Since the piston moves smoothly in the cylinder, the only forces are those shown:

$$\Sigma F = 0 \Rightarrow pA = P_{\text{atm}}A + \text{Weight}$$

$$\therefore p = P_{\text{atm}} + \frac{\text{Weight}}{A}$$

$$= 1 \text{ bar} + \left(\frac{471.1 \text{ N}}{3.14 \times 10^{-4} \text{ m}^2} \right) \left(\frac{1 \text{ bar}}{10^5 \text{ N/m}^2} \right) = 16 \text{ bar.}$$

$$A = \frac{\pi D^2}{4} = \frac{\pi (0.02 \text{ m})^2}{4} = 3.14 \times 10^{-4} \text{ m}^2$$

The process is sketched on the accompanying p-v diagram. Since pressure is constant, we get

$$\frac{W}{m} = p(v_2 - v_1)$$

$$= (16 \text{ bar}) \left(\frac{10^5 \text{ N/m}^2}{1 \text{ bar}} \right) (0.02017 - 0.0121) \frac{\text{m}^3}{\text{kg}} \left(\frac{1 \text{ kJ}}{10^3 \text{ N} \cdot \text{m}} \right) = 12.9 \frac{\text{kJ}}{\text{kg}} \leftarrow$$

PROBLEM 3.35

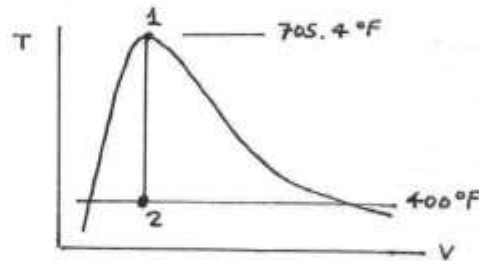
KNOWN: A closed, rigid tank is filled with water, initially at the critical point. The water is cooled to a given temperature

FIND: For the water, show the process on a T-v diagram and determine the heat transfer, in Btu/lb.

SCHEMATIC & GIVEN DATA:



Initially at the critical point.
Finally at 400°F



ENGINEERING MODEL

1. The water in the tank is the closed system.
2. The only energy transfer is by heat.
3. Kinetic and potential energy effects can be ignored.

ANALYSIS:

Since the total mass and total volume remain constant, the water undergoes a constant-volume process, as shown in the T-v diagram.

With 2 and 3, the energy balance reduces as follows:

$$\Delta U + \Delta KE + \Delta PE = Q - W \Rightarrow Q = \Delta U = m(u_2 - u_1), \text{ or}$$

$$\frac{Q}{m} = u_2 - u_1 \quad (1)$$

From Table A-2E, $u_1 = 872.6 \text{ Btu/lb}$

To find u_2 , first evaluate x_2 using $v_2 = v_1 = 0.05053 \text{ ft}^3/\text{lb}$.

$$x_2 = \frac{v_2 - v_f}{v_g - v_f} = \frac{0.05053 - 0.01864}{1.866 - 0.01864} = 0.0173 \quad (1.73\%)$$

$$\Rightarrow u_2 = u_f + x_2(u_g - u_f) = 374.3 + (0.0173)(1116.6 - 374.3) = 387.1 \text{ Btu/lb},$$

where u_f and u_g are from Table A-2 at 400°F.

Substituting values into Eq. (1),

$$\frac{Q}{m} = (387.1 - 872.6) \frac{\text{Btu}}{\text{lb}} = -485.5 \text{ Btu/lb}$$

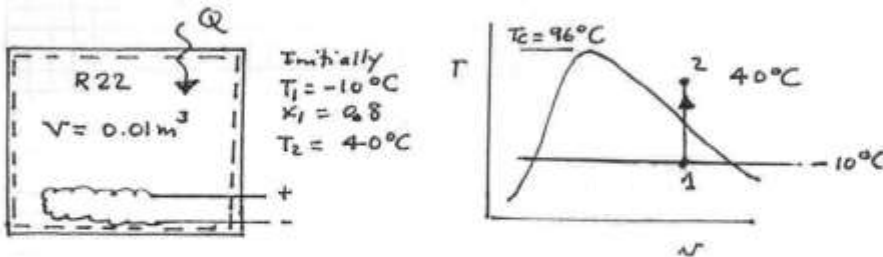
Energy transfer by heat from the water.

PROBLEM 3.44

KNOWN: A closed, rigid tank fitted with a resistor is filled with Refrigerant 22, initially a two-phase liquid-vapor mixture. State data and operating data are provided.

FIND: Determine the heat transfer from the refrigerant, in kJ.

SCHEMATIC & GIVEN DATA:



ENGINEERING MODEL:

1. The closed system is the R22 plus resistor.
2. The mass and volume of the fine-wire resistor are negligible.
3. Kinetic and potential energy can be ignored.

ANALYSIS:

Since the total mass and volume of the R22 remain constant, the refrigerant undergoes a constant specific volume process, as shown in the T-v diagram.

An energy balance reduces as follows:

$$(\Delta U)_{\text{R22}} + \underbrace{\Delta U_{\text{Resistor}}^0}_{\text{by assumption 2}} + \cancel{\Delta KE} + \cancel{\Delta PE} = Q - W$$

$$\Rightarrow Q = \Delta U_{\text{R22}} + W = m(u_2 - u_1) + W \quad (1)$$

With data from Table A-7,

$$\begin{cases} u_1 = u_f + x_1(u_g - u_f) = 33.27 + 0.8(223.02 - 33.27) = 185.07 \text{ kJ/kg} \\ v_1 = v_f + x_1(v_g - v_f) = (0.7606/10^3) + 0.8(0.0652 - (0.7606/10^3)) = 0.0523 \text{ m}^3/\text{kg} \end{cases}$$

Interpolating in Table A-9 with $T_2 = 40^\circ\text{C}$ and $v_2 = v_1$, $u_2 = 250.33 \text{ kJ/kg}$

The mass is $m = \frac{V}{v_1} = \frac{0.01 \text{ m}^3}{0.0523 \text{ m}^3/\text{kg}} = 0.191 \text{ kg}$

With Eq. 2.21, $W = -\int i \Delta t$

$$= -(12 \text{ V})(5 \text{ amp})(5 \text{ min}) \left| \frac{1 \text{ Watt/amp}}{1 \text{ volt}} \right| \left| \frac{60 \text{ s}}{1 \text{ min}} \right| \left| \frac{1 \text{ J/s}}{1 \text{ W}} \right| \left| \frac{1 \text{ kJ}}{10^3 \text{ J}} \right|$$

$$= -18 \text{ kJ}$$

Collecting results, Eq. (1) gives

$$Q = (0.191 \text{ kg})(250.33 - 185.07) \frac{\text{kJ}}{\text{kg}} + (-18 \text{ kJ}) = -5.54 \text{ kJ}$$

With assumption 2, we associate this heat transfer with the R22. Accordingly, the heat transfer from the refrigerant is 5.54 kJ.